

Metallurgy of Iron

Iron:-

Iron (Fe) is a chemical element with the atomic no. 26. This means that every iron atom has 26 protons in its nucleus.

→ Iron is silvery-white, lustrous metal that is ductile and malleable, especially when hot. It is a good conductor of heat and electricity and is ferromagnetic.

→ Iron is chemically active. It readily combines with oxygen in the presence of moisture to form iron (III) oxide.

Occurrence of Iron:-

Iron is present in the Earth's crust in the form of oxides, carbonates and sulphides.

1- Iron Pyrite (FeS_2) is widely distributed.

but is not suitable for production of metallic iron and steel. The reason is that it is almost impossible to remove the last traces of sulphur. The presence of sulfur makes the resulting steel too brittle to be useful.

2- Siderite (FeCO_3) is a valuable iron mineral that can be converted to iron oxide by heating.

3- Hematite (Fe_2O_3) and Magnetite (Fe_3O_4 , really $\text{FeO} \cdot \text{Fe}_2\text{O}_3$) are the most abundant iron oxide minerals.

4- Taconite Ores contains iron oxides mixed with silicates and are more difficult to process than the others. However, taconite ores are being increasingly used as the most desirable ores are consumed.

Concentration of ore:-

To concentrate the iron in iron ores, we take the advantage of the natural magnetism of Fe_3O_4 hence its name, magnetic.

The (Fe_3O_4) particles can be separated from the **gangue** by magnets. The ores that are not magnetic are often converted to (Fe_3O_4) . Hematite (Fe_2O_3) is partially reduced to magnetite (Fe_3O_4) . Siderite $(FeCO_3)$ is first converted to FeO thermally, then oxidized to Fe_2O_3 and finally reduced to (Fe_3O_4) .

Reactions:-



Sometimes the non-magnetic ores are concentrated by floatation process.

Extraction of Iron from its Ores:-

Iron is extracted from its oxide ores especially from the (magnetite, haematite and lime ores).

There are some various steps for extraction of iron:-

(I)

Concentration of Ore:-

The concentration of ore is done by (gravity process). The ore is crushed into small pieces. Then it is washed with water to remove silicious impurities. The washed ore is then subjected to (Electromagnetic Separation.)

(II)

Calcination and Roasting:-

The concentrated ore is heating excess of air. During heating following reactions takes place:-

(a) Removeable of moisture and CO_2 .

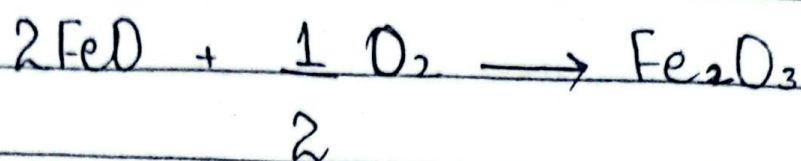


(b) Oxidation of sulphur, arsenic etc. and their removal as volatile gases.



(c) Oxidation of Ferrous oxide to

ferric oxide.



(d) The entire mass becomes porous.

(III)

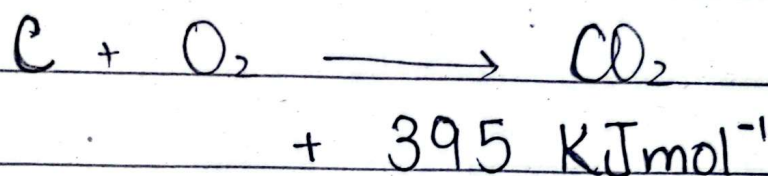
Smelting:-

The calcined ore is mixed with limestone (CaCO_3) and coke in the 8 : 1 : 4 and introduced in a blast furnace for smelting.

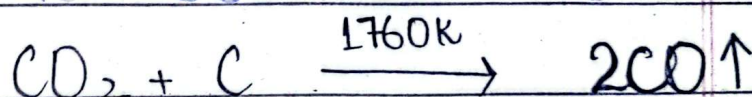
The whole material is called **Charge**. This charge is introduced in the furnace lowering the cup and cone arrangement. At the same time the furnace is lit (H_2 , H_2S) and a blast of hot air is sent upwards through the tuyeres. The temp. varies from **1800 to 525 K** in the furnace from bottom to top. On the basis of variation in temperature, there are four zones with different chemical changes occur:-

(a) Combustion Zone:-

This is the lowest part of the furnace above hearth where the temp. is about **1760-1860 K**. In this zone carbon burns in the presence of hot air producing CO_2 . A lot of heat is also produced during burning of carbon.



When CO_2 raises upwards, it meets with red hot coke. Here it is reduced to CO .



(b) Reduction Zone:-

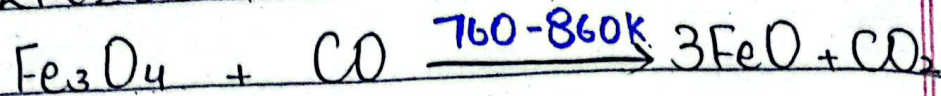
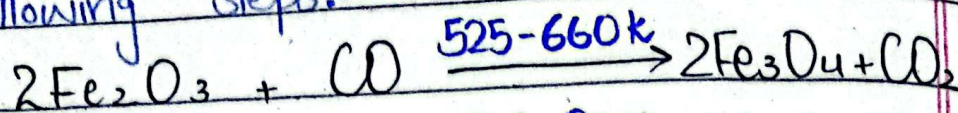
The uppermost part of the furnace is responsible for reduction.

The temperature varies from **520 to 960 K**. Iron formed is called

spongy iron. Here oxide ore is reduced to iron in this zone.

The reaction takes place through

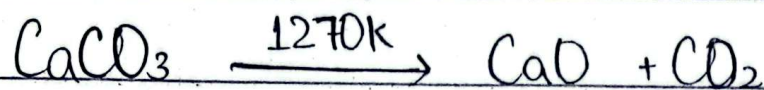
Following steps:-



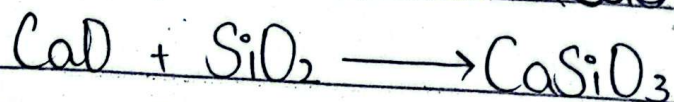
(c) Slag Formation Zone:-

The formation of slag is takes place in central zone. Here the temperature varies from **1070-1270K**

The limestone present in the charge decomposes into (calcium oxide and CO_2).



The calcium oxides so produced is acts as a **flux**. It combines with silica present as an impurity or gangue, to form a fusible slag of calcium silicate (CaSiO_3)



(gangue) Silica

Slag

Silicates, phosphates and Manganates present as impurities in the ore are reduced to (Si, P and Mn) respectively.

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(D) Zone of Fusion.

The fusion zone is just above the zone of combustion.

Here temp. ranges b/w **1460-1760K**

The spongy iron which has already absorbed **(C, Si, P, Mn)** etc. melts

at **1570K**. It is collected at

the bottom of the hearth. (The

slag being lighter floats over the molten iron.) It prevents the oxidation

of molten metal. The slag and molten metal are removed from their respective holes.

The molten metal is run into the moulds and is allowed

to solidify. Iron obt. from the

blast furnace is called **Pig Iron**.

It contains **(93% iron; 5%**

carbon) The rest of silicon, manganese,

and phosphorous etc are present

as impurities. The pig iron is re-melted

and cast or poured into mould. The

product so obt. is known as **(cast iron)**

Varieties of Iron

- Cast Iron
- Wrought Iron

Cast Iron

Wrought Iron

Define

It is the most impure form of iron. It contains high % of carbon, 2.5-5% and 2% of impurities like Si, P, Mn and S.

It is the purest form of iron. It contains the lowest percentage of carbon, from 0.1 - 0.25%.

Structure

Crystalline Structure.

In gray cast iron, due to the presence of Carbon present as interconnected graphite flakes. In white cast iron, Carbon is

Unique fibrous structure

of slag inclusions elongated during the working process.

Chemically bonded.

Strength

High Compressive Strength.

High tensile and compressive strength.

Ductility

Very low
brittle

High
can be drawn into wires

Malleability

Non-Malleable

Highly malleable

Hardness

Generally high

Softer than
cast iron

Weldability

difficult to weld

Easier to weld

Corrosion Resistance

Moderate resistance

Good Corrosion resistance

Fatigue Resistance

Lower

Generally high.

Applications

Its used for:-

- Engine blocks and cylinder heads
- Pipes and fittings
- Manhole covers and grates
- Certain cookware
- Decorative elements

It's suitable for:-

- Ornamental iron work like gates, railings.
- Historically for anchors & chains
- Nuts, bolts and rivets
- Durable Cookware
- Handrails and balustrades.

Manufacture of Cast Iron:-

Cast iron is formed by melting pig iron (the product of smelting iron ore) along with scrap and sometimes other materials in a furnace (like a cupola or electric furnace). The molten iron, relatively high carbon content (2-4%) is then poured into molds to solidify into desired shape.

The cooling rate during solidification determines the type of cast iron formed (eg gray or white iron).

After cooling the casting is removed and finished.

Manufacture of Wrought Iron:-

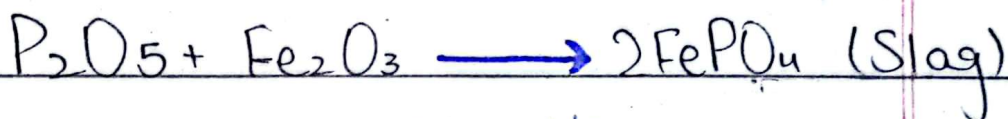
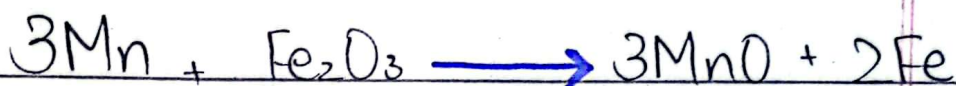
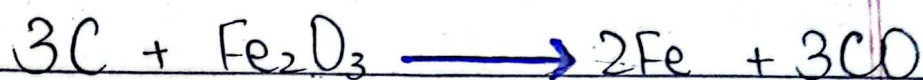
A special type of reverberatory furnace called puddling furnace is used. The hearth of this furnace is lined with haematite (Fe_2O_3). The cast iron is melted on the hearth by the hot gases and stirred

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with long iron rods. The impurities of cast iron are oxidized by oxygen of haemetite (lining). Oxides of carbon and sulphur being volatile escape while those of Mn, P, Si form slag.

Reaction:-



When impurities are removed, the melting point of the metal rises. It becomes a semi-solid mass. The semi-solid mass is taken out and is beaten under steam hammers. This is done to remove as much as of slag as possible. In this way we almost get pure iron known as wrought Iron.



THE END!

